

10.13 DIMENSIONAL REGULARIZATION AND THE MS SCHEME

THE CORE MECHANISM OF DIMENSIONAL REGULARIZATION

In dimensional regularization (dim reg), we analytically continue the spacetime dimension from $d = 4$ to $d = 4 - \epsilon$ (some books use $d = 4 - 2\epsilon$; it is a matter of convention). The fundamental quantity governing divergences is the Euler Gamma function, as seen in the master formula for Euclidean integrals:

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + \Delta)^n} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2} \quad (10.13.1)$$

The Gamma function $\Gamma(z)$ has simple poles at non-positive integers $z = 0, -1, -2, \dots$. Therefore, divergences (as poles in ϵ) appear when the argument $n - d/2$ approaches one of these integers. For one-loop integrals of this form the Gamma-function poles are the *only* source of ultraviolet divergences: if $\Gamma(n - d/2)$ is finite, the integral is finite in this scheme.

ODD VERSUS EVEN DIMENSIONS

The behavior of loop integrals differs between odd and even dimensions. For a one-loop diagram ($L = 1$) with I internal scalar propagators and no derivative couplings, the superficial degree of divergence is

$$\omega = dL - 2I = d - 2I.$$

Look at its parity:

- If d is even, $\omega = (\text{even}) - 2I$ is even, so $\omega = 0$ (a logarithmic divergence) is possible.
- If d is odd, $\omega = (\text{odd}) - 2I$ is *always* odd.

So, by power counting, a one-loop diagram in an odd dimension cannot be logarithmically divergent. This explains the “special” nature of odd dimensions in dim reg:

- **Even dimensions** ($d = 2, 4, 6, \dots$): $d/2$ is an integer. One-loop diagrams can have $\omega = 0, 2, 4, \dots$. The Gamma-function argument $n - d/2$ can hit a pole, producing $1/\epsilon$ divergences for *both* power-law and logarithmic cases.

- **Odd dimensions** ($d = 3, 5, 7, \dots$): $d/2$ is a half-integer. One-loop diagrams can only have $\omega = 1, 3, 5, \dots$. The Gamma-function argument $n - d/2$ is *always* a half-integer, which is *never* a pole. Thus **all one-loop diagrams in odd dimensions are finite in dim reg**. But we know a divergence is a divergence; it does not disappear. Dim reg does something special, explained below.

Clarification: this only applies at one loop. In odd dimensions, logarithmic divergences appear at even loop orders, where $\omega = Ld - 2I$ can vanish. In ϕ^6 theory in $d = 3$, the one-loop diagrams are finite, but at two loops $\omega = 2d - 2I = 0$ for $I = 3$, and dim reg shows the corresponding $1/\epsilon$ pole. This is why renormalization is still needed in odd dimensions.

POWER-LAW VERSUS LOGARITHMIC DIVERGENCES

Dim reg treats the two kinds of divergence very differently.

- **Scaleless integrals**, which have no mass parameter at all, like $\int d^d k (k^2)^a$, are analytically continued to zero. This is a built-in feature of the scheme. Introducing the renormalization scale μ ($d^d k \rightarrow \mu^{4-d} d^d k$, to keep couplings dimensionless) does not change this: it cannot generate a mass scale inside the integral.
- **Integrals with a mass scale** $\Delta = m^2$ keep only the part proportional to powers of m .

In $d = 4 - \epsilon$:

- **Logarithmic** ($\omega = 0$):

$$I_{\log} = \int \frac{d^d k}{(k^2 + \Delta)^2} \propto \Gamma(2 - d/2) = \Gamma(\epsilon/2) = \frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon).$$
This is a “pure” $1/\epsilon$ pole.
- **Power law** ($\omega = 2$): $I_{\text{quad}} = \int \frac{d^d k}{k^2 + \Delta} \propto \Gamma(1 - d/2) = \Gamma(-1 + \epsilon/2) \sim m^2 \times \frac{1}{\epsilon}$.
The m^2 comes from the factor $(1/\Delta)^{n-d/2}$ in Eq. (10.13.1), which becomes $\Delta^{d/2-1} \approx \Delta = m^2$.

What does the m^2/ϵ pole correspond to? Compare a Euclidean momentum cutoff Λ :

$$\begin{aligned} \int^\Lambda \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m^2} &= \frac{1}{16\pi^2} \left[\Lambda^2 - m^2 \ln \frac{\Lambda^2}{m^2} + \mathcal{O}\left(\frac{m^4}{\Lambda^2}\right) \right] \quad (\text{cutoff}), \\ \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + m^2} &= -\frac{m^2}{16\pi^2} \left[\frac{2}{\epsilon} + \dots \right] \quad (\text{dim reg}). \end{aligned}$$

The quadratically divergent integral contains a subleading *logarithmic* divergence, $-m^2 \ln \Lambda^2$, and that is what the pole is: $\ln \Lambda^2 \leftrightarrow 2/\epsilon$. The pure power Λ^2 has no counterpart in dim reg; it is simply dropped. So it is tempting to say that in even dimensions dim reg “tags” the pole with the

mass dimension of the power-law divergence, but more precisely, **every** $1/\epsilon$ **pole is a logarithmic divergence**, multiplied by whatever powers of the masses the dimensions require. In odd dimensions there is no logarithm at one loop, and a power-divergent integral is evaluated to a finite multiple of m^ω .

Dim reg is subtle: it discards every pure power-law divergence. For a scaleful integral (with masses in Δ) what is left of a divergence of degree ω is m^ω times a $1/\epsilon$ pole (a logarithm) in even dimensions, or times a finite number in odd dimensions. For a scaleless integral nothing is left. This automatic subtraction of pure power divergences is both a feature (computational simplicity) and a limitation (it hides the sensitivity of the results to high scales).

1966 THE $\overline{\text{MS}}$ SCHEME: WHAT ARE WE REALLY DOING?

1967 In the $\overline{\text{MS}}$ (modified minimal subtraction) scheme, we:

- 1968 1. Subtract **only** the $1/\epsilon$ poles (together with the constants $\gamma_E - \ln(4\pi)$ that accompany them).
- 1970 2. Do *not* “ignore” power-law divergences. Rather, we **implicitly**
- 1971 **accept dim reg’s evaluation of them**, finite or zero.

1972 This works because in a renormalizable QFT all power-law divergences can be absorbed by the same counterterms that absorb the logarithmic ones. The $\overline{\text{MS}}$ scheme appears “minimal” only because the regulator has already done the heavy lifting of removing $\Lambda^2, \Lambda^3, \dots$

Contrast with Pauli–Villars regularization:

$$\Pi_{\text{PV}}(p^2) \sim c_1 \Lambda^2 + c_2 m^2 \ln \left(\frac{\Lambda^2}{\mu^2} \right) + \text{finite}.$$

1976 Here we explicitly see both quadratic (Λ^2) and logarithmic ($\ln \Lambda^2$) divergences. A counterterm must be “brute-forced” to cancel the Λ^2 term explicitly. In $\overline{\text{MS}}$, the Λ^2 term never appears.

1979 ONLY THE $1/\epsilon$ POLES DRIVE THE RUNNING

The renormalization-group flow, described by the β -function $\beta(g_R) = \mu dg_R/d\mu$, tells us how a coupling g_R must change with the scale μ to keep physical observables independent of our choice of μ . The bare coupling g_0 is independent of μ . In $d = 4 - \epsilon$ dimensions,

$$g_0 = \mu^\epsilon \left(g_R + \frac{a_1(g_R)}{\epsilon} + \dots \right).$$

Setting $\mu dg_0/d\mu = 0$ and keeping the ϵ^0 terms gives

$$\beta(g_R) = \left(g_R \frac{d}{dg_R} - 1 \right) a_1(g_R),$$

1980 so β is determined *entirely* by a_1 , the residue of the $1/\epsilon$ pole. For $\lambda\phi^4/4!$
 1981 theory, $a_1 = 3\lambda^2/16\pi^2$ and $\beta = 3\lambda^2/16\pi^2$.

1982 The same holds for the mass. The m^2/ϵ pole of the quadratically divergent
 1983 tadpole is a genuine $1/\epsilon$ residue, and it drives the running of m^2 : in $\lambda\phi^4/4!$
 1984 theory, $m_0^2 = m^2(1 + \lambda/16\pi^2\epsilon + \dots)$, so that $\mu dm^2/d\mu = \lambda m^2/16\pi^2$ at one loop.
 1985 What does *not* run is the pure power Λ^2 , which dim reg has already
 1986 discarded. So the precise statement is: the running is fixed by the
 1987 logarithmic divergences, including the logarithms that sit inside
 1988 power-divergent integrals with mass-dimension prefactors. The finite
 1989 parts left after subtraction are not μ -independent: they carry $\ln(\mu^2/m^2)$
 1990 and $\ln(\mu^2/p^2)$, and their μ -dependence is exactly what the running of the
 1991 couplings compensates.

1992 MASS INDEPENDENCE, THRESHOLDS AND DECOUPLING

1993 Because $\overline{\text{MS}}$ subtracts only the $1/\epsilon$ poles, and their coefficients are
 1994 polynomials in the masses with mass-independent coefficients, the
 1995 renormalization factors Z (and the counterterm coefficients $\delta_m, \delta_\lambda, \dots$) do
 1996 not depend on the masses. The β -functions of the dimensionless
 1997 couplings are therefore mass independent: the scheme is “blind” to the
 1998 masses of the particles in the loops.

1999 The physical consequence is that $\overline{\text{MS}}$ **does not see particle thresholds**.
 2000 When the scale drops below the mass M of a heavy particle, that particle
 2001 should stop affecting low-energy physics (the Appelquist–Carazzone
 2002 decoupling theorem, 1975). In a physical, momentum-subtraction
 2003 scheme this happens automatically: the heavy particle’s contribution to
 2004 the running of the light couplings switches off for $\mu \ll M$. In $\overline{\text{MS}}$ the heavy
 2005 particle keeps contributing to the β -functions at all μ , and its effect at low
 2006 energy reappears as large logarithms $\ln(M/\mu)$ in the finite parts. One
 2007 restores decoupling by hand: integrate the heavy particle out at $\mu \sim M$
 2008 and match the full theory to an effective field theory without it.

2009 **Example:** in the Standard Model the top quark ($m_t \approx 173$ GeV) should not
 2010 affect the running of the light couplings at $\mu \ll m_t$. In $\overline{\text{MS}}$ this has to be
 2011 implemented through decoupling relations: for instance, the strong
 2012 coupling below m_t is run with five quark flavors after matching at $\mu \sim m_t$.

2013 PHYSICAL IMPLICATIONS AND THE LIMITS OF 2014 MATHEMATICAL ELEGANCE

2015 While dim reg + $\overline{\text{MS}}$ is computationally optimal, it obscures important
 2016 physical puzzles by “hiding” power-law divergences.

2017 THE HIERARCHY PROBLEM

In cutoff regularization, the top-quark loop corrects the Higgs mass by

$$\delta m_H^2 \sim y_t^2 \Lambda^2 + y_t^2 m_t^2 \ln \left(\frac{\Lambda^2}{\mu^2} \right).$$

The quadratic term Λ^2 represents a dramatic fine-tuning problem if $\Lambda \sim M_{\text{Pl}}$. In dim reg,

$$\delta m_H^2 \sim y_t^2 m_t^2 \left(\frac{1}{\epsilon} + \text{finite} \right).$$

2018 The Λ^2 divergence is gone. The fine-tuning problem becomes
2019 mathematically invisible, but physically it remains: if the Higgs couples
2020 to any heavy particle of mass M , dim reg gives a correction proportional
2021 to M^2 , and the sensitivity to the heavy scale is back.

2022 THE COSMOLOGICAL CONSTANT PROBLEM

With a cutoff, the zero-point energy density of a free field is

$$\rho_{\text{vac}} = \frac{1}{2} \int^\Lambda \frac{d^3 k}{(2\pi)^3} \sqrt{\mathbf{k}^2 + m^2} \sim \frac{\Lambda^4}{16\pi^2}.$$

2023 In dim reg, for a massless field this is a scaleless integral, which the rules
2024 define to be **zero**; for a massive field only a term proportional to
2025 $m^4 \ln(m^2/\mu^2)$ survives. This regularization *deletes* the Λ^4 from the
2026 calculation, but it does not remove the problem: the m^4 contributions of
2027 the known particles alone exceed the observed vacuum energy by many
2028 orders of magnitude.

2029 THE LESSON

2030 (As I emphasized in class.) We must distinguish between mathematical
2031 convenience and physical insight. Different (reasonable) regularization
2032 schemes make different problems or properties of the underlying theory
2033 apparent. Dim reg gives the correct answers for renormalized observable
2034 quantities (like scattering amplitudes) but may obscure the deeper
2035 physical questions about naturalness, UV sensitivity, and decoupling
2036 that are made manifest in cutoff-based schemes. The choice of
2037 regularization scheme is not just computational: it embodies a
2038 philosophy about what constitutes “real” physics. For a bottom-up
2039 effective field theorist, a renormalizable theory is self-consistent to
2040 arbitrarily high scales, and won’t see any problem with the underlying
2041 theory. For a top-down UV theorist who wants to understand the
2042 ultimate Lagrangian of nature, consistency is only one aspect;
2043 calculability (and hence the predictivity of the bare parameters) is crucial,
2044 as long as one does not want to admit that we live in a special and

2045 randomly tuned universe. Misinterpretations of the $\overline{\text{MS}}$ scheme have
2046 caused a tremendous amount of confusion in the field, even today. I hope
2047 you learned something from this note.

2048 SUMMARY

- 2049 • **Mathematical advantage:** $\text{dim reg} + \overline{\text{MS}}$ is the most efficient
2050 scheme, preserving symmetries and keeping only the universal
2051 logarithmic running.
- 2052 • **Odd dimensions** are a clean laboratory: one-loop finiteness by
2053 power counting, and $1/\epsilon$ poles only at even loop orders ($L = 2, 4, \dots$)
2054 in scalar theories.
- 2055 • **Every $1/\epsilon$ pole is a logarithm.** Pure power divergences are
2056 discarded, and the m^2/ϵ of a quadratically divergent integral is the
2057 logarithm inside it; it drives the running of m^2 .
- 2058 • **Physical limitation:** the scheme makes fine-tuning problems (like
2059 the hierarchy problem) mathematically invisible, sets scaleless
2060 power divergences (like the massless vacuum energy) to zero, and,
2061 being mass independent, does not decouple heavy particles by
2062 itself.